Journal of Computational and Applied Mathematics 11 (1984) 307–314
North-Holland

Bifurcation studies in reaction-diffusion II

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Received 12 December 1983
Revised 5 June 1984

Keywords: Reaction diffusion, bifurcation, pseudo-spectral method.

1. Introduction

The bifurcation of steady state solutions of reaction-diffusion equations has been the subject of interest in recent times [1,2,4,5,9]. In an earlier paper [6] we studied the reaction-diffusion equation

$$u_t = u_{xx} + u(1-u)(u-a), \quad 0 < a < \tfrac{1}{2} \tag{1.1}$$

subject to the boundary conditions

$$u(-L, t) = u(L, t) = b, \quad b \text{ constant} \tag{1.2}$$

and the initial condition

$$u(x, 0) = u_0(x),$$

and estimated the critical lengths of the domain at which bifurcation occurs in the cases $b = 0$, a and 1. Also, we were able to characterise some of the non constant steady state solutions which appear beyond the points of instability by a steady state solution of the form

$$u(x) = 3a\left\{\sqrt{(2-a)(\tfrac{1}{2}-a)}\,\cosh(\sqrt{a}\,x) + (1+a)\right\}^{-1}. \tag{1.3}$$

As a follow up to this earlier paper [6], we now study in detail the solutions bifurcating from the equilibrium state $u = a$. To simplify matters, we use the transformation

$$x = 2L\left(x^1 - \tfrac{1}{2}\right), \qquad t = 4L^2 t^1 \quad \text{and} \quad V = u - b.$$

Omitting the primes, the transformed problem is

$$V_t = V_{xx} + 4L^2 g(V), \quad 0 < a < \tfrac{1}{2}, \tag{1.4}$$

subject to the boundary conditions

$$V(0, t) = V(1, t) = 0 \tag{1.5}$$

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and the initial condition

$$V(x,0)=u_0(x)-b,$$

where

$$g(V)=-V^3+(1-2a)V^2+a(1-a)V, \tag{1.6}$$

and the solution corresponding to (1.3) is now given by

$$V=3a\left\{\sqrt{(2-a)(\tfrac{1}{2}-a)}\ \cosh(\sqrt{a}\,L(2x-1))+(1+a)\right\}^{-1}-a. \tag{1.7}$$

2. Linear analysis

Let us introduce a perturbation v to the constant solution $V=0$ (i.e. $u=a$). Then, the linearised equation for the perturbation v is given by

$$v_t=v_{xx}+\alpha\{a(1-a)v\},\quad \alpha=4L^2, \tag{2.1}$$

together with the boundary conditions

$$v(0,t)=v(1,t)=0 \tag{2.2}$$

where the constant state $V=0$ is stable if the eigenvalues $\mu(\alpha)$ of the operator

$$\mathscr{L}\equiv\left[\frac{\partial^2}{\partial x^2}+\alpha a(1-a)\right]$$

are negative. i.e., if

$$\mu(\alpha)=-n^2\pi^2+\alpha a(1-a)<0,\quad n=1,2,\ldots.$$

Which implies that instability occurs when

$$\alpha=\pi^2/a(1-a)\quad(=\alpha_B,\text{ say}) \tag{2.3}$$

and the eigenfunction of $\mathscr{L}$ corresponding to this critical mode is

$$v=c\sin\pi x. \tag{2.4}$$

Note that the estimate obtained for the critical value of α in [6] is $4\pi^2$, which is the value obtained from $\min_{0<a\leq 1/2}\{\alpha_B\}$.

3. Nonlinear analysis

Now we will show that bifurcation occurs beyond the critical value given by (2.3) and obtain the approximate steady state solutions in the neighbourhood of the bifurcation point. Let us re-write equation (2.1) (for the steady state), with the inclusion of the nonlinear terms as

$$G(\alpha,v)\equiv\mathscr{L}_B v+(\alpha-\alpha_B)a(1-a)v-\alpha\{v^3-(1-2a)v^2\}=0 \tag{3.1}$$

where

$$\mathscr{L}_B = \frac{\partial^2}{\partial x^2} + \alpha_B a(1-a).$$

Since, $\mu'(\alpha_B) \neq 0$, there is a non-trivial branch (α, v) which may be expressed as $v = v(\varepsilon)$, $\alpha = \alpha(\varepsilon)$, where $v(0) = 0$, $\alpha(0) = \alpha_B$; the functions v and α being power series in a new variable ε, convergent for sufficiently small $|\varepsilon|$, [8]. Also, $\mu'(\alpha_B) \neq 0$ guarantees that the bifurcation consists of a simple curve of non-trivial solutions (in a general case, there may be several solution curves intersecting at the bifurcation point). Let us assume the power series expansions to be

$$v = \varepsilon v_0 + \varepsilon^2 v_1 + \varepsilon^3 v_2 + \cdots \quad \text{and} \quad \alpha - \alpha_B = \varepsilon\gamma_1 + \varepsilon^2\gamma_2 + \varepsilon^3\gamma_3 + \cdots. \tag{3.2}$$

Introducing these expansions into (3.1) and identifying the terms with equal powers of ε, we get

$$\mathscr{L}_B v_k = a_k, \quad k = 0, 1, 2, \ldots \tag{3.3}$$

with

$$v_k(0) = v_k(1) = 0,$$

where

$$a_0 = 0, \qquad a_1 = -(1-2a)\alpha v_0^2 - a(1-a)\gamma_1 v_0,$$
$$a_2 = \alpha v_0^3 - 2\alpha(1-2a)v_0 v_1 - a(1-a)(\gamma_1 v_1 + \gamma_2 v_0).$$

Since $\mathscr{L}_B$ has zero as a simple eigenvalue, (3.3) has a solution if and only if a_k $(k = 0, 1, 2, \ldots)$ satisfies the solvability condition given by

$$\int_0^1 a_k \sin \pi x \, dx = 0. \tag{3.4}$$

Now, let us look at the solutions of (3.3). It follows immediately that

$$v_0 = c \sin \pi x \tag{3.5}$$

and the solvability condition (3.4) implies that

$$\gamma_{1/c} = -8\alpha(1-2a)/\{3\pi a(1-a)\} \quad (= h_1, \text{ say}). \tag{3.6}$$

To solve

$$\mathscr{L}_B v_1 = a_k,$$

we assume that v_1 has a Fourier series expansion

$$v_1 = \sum_{l=1}^{\infty} p_l \sin l\pi x \quad \Rightarrow \quad \{-l^2\pi^2 + \alpha_B a(1-a)\} p_l = b_l \tag{3.7}$$

where

$$b_l = \begin{cases} 8(1-2a)\alpha c^2/\pi l(l^2-4), & l \neq 1, \text{odd}, \\ 0, & l = 1 \text{ or } l \text{ is even}. \end{cases}$$

Hence,

$$v_1 = \sum_{\substack{l \text{ is odd} \\ l \neq 1}} p_l \sin l\pi x \tag{3.8}$$

where

$$p_l = -8\alpha(1-2a)c^2/\pi^3 l(l^2-1)(l^2-4), \quad l \neq 1, \text{odd}.$$

Again, using the solvability condition (3.4) we obtain

$$\gamma_2/c^2 = \frac{2}{a(1-a)}\left\{\frac{3}{8}\alpha - \frac{64}{\pi^4}\alpha^2(1-2a)^2 S\right\} \quad (= h_2, \text{ say}) \tag{3.9}$$

where

$$S = \sum_{\substack{l \text{ is odd} \\ l \neq 1}} \frac{1}{l^2(l^2-4)^2(l^2-1)} = \left(\frac{7}{108} - \frac{5\pi^2}{768}\right).$$

Substituting (3.6) and (3.9) into (3.2) we get

$$\alpha - \alpha_B = \varepsilon\gamma_1 + \varepsilon^2\gamma_2 = (\varepsilon c)h_1 + (\varepsilon c)^2 h_2 \tag{3.10}$$

and

$$v = (\varepsilon c)\sin \pi x + (\varepsilon c)^2 \sum_{l \neq 1, \text{odd}} \frac{p_l}{c^2}\sin l\pi x. \tag{3.11}$$

From (3.10), we have

$$\varepsilon c = \left\{-h_1 \pm \left(h_1^2 + 4h_2(\alpha - \alpha_B)\right)^{1/2}\right\}/2h_2. \tag{3.12}$$

This means two new solutions appear for α above and below the critical value α_B. However, when $\alpha < \alpha_B$ in order to get real values in (3.12) we should have

$$\alpha \geqslant \alpha_{\min} = \frac{\beta_2}{2\beta_1}\left\{1 - \left(\frac{1 - 4\beta_1\beta_3}{\beta_2^2}\right)^{1/2}\right\}. \tag{3.13}$$

where

$$\beta_1 = \frac{64}{\pi^4}(1-2a)^2 S, \quad \beta_2 = \frac{8}{9\pi^2}\left\{\frac{(1-2a)^2}{a(1-a)}\right\} + \frac{64}{\pi^4}(1-2a)^2 S\alpha_B + \frac{3}{8}, \quad \beta_3 = \frac{3}{8}\alpha_B.$$

It should be noted that the use of power series expansions in ε, is valid only if $G(\alpha, v)$ is analytic in (α, v) in a neighbourhood of $(\alpha_B, 0)$. Also, it should be pointed out that for gradient driven reaction-diffusion equations (i.e. reaction-diffusion equations containing functions of $|\nabla u|^2$ in the nonlinearities), the above perturbation expansion method may be unsatisfactory [8].

In the following section we do a numerical study of the problem (1.4) which determines which of these new bifurcating solutions are asymptotically stable.

4. Numerical study

Considering the analysis carried out in the previous section, the pseudo-spectral method seems to be a suitable method for carrying out numerical calculations of the equation (1.4) together with the appropriate boundary and initial conditions. In this method we approximate the

solution by

$$V(x, t) \approx \tilde{V}(x, t) = \sum_{i=1}^{m} c_i(t)\psi_i(x) \tag{4.1}$$

which is known as the Kantorovich (semi-discrete) method (where, the m-dimensional approximating subspace is chosen to contain functions of x only, satisfying the boundary conditions) and put the residual $r(x, t)$ given by

$$r(x, t) = \tilde{V}_t - \tilde{V}_{xx} - \alpha g(\tilde{V}) \tag{4.2}$$

equal to zero at selected points $x_1, x_2, \ldots, x_m$ in the solution domain. The basis functions $\psi_i(x)$ in (4.1), are chosen to be eigenfunctions of the Laplacian and for the present study equally spaced points in the solution domain are chosen to be the collocation points x_i, $i = 1, 2, \ldots, m$. To avoid nonlinearity in the resulting algebraic problem following [3] we implement a predictor–corrector formulation given by

$$P\colon\ \tilde{V}_p(x_i, t_{k+1}) = \tilde{V}(x_i, t_k) - \Delta t \sum_{j=1}^{m} \lambda_j C_j(t_k)\phi_{ij} + \Delta t \alpha g(x_i, t_k) \tag{4.3}$$

(Δt, time step, $\phi_{ij} = \psi_j(x_i)$ and λ_j, the eigenvalues of the Laplacian).

$$\begin{aligned} C\colon\ \tilde{V}(x_i, t_{k+1}) = \tilde{V}(x_i, t_k) &- \tfrac{1}{2}\Delta t \sum_{j=1}^{m} \lambda_j \phi_{ij}\{C_j(t_k) + C_j^p(t_{k+1})\} \\ &+ \tfrac{1}{2}\Delta t \alpha\{g(x_i, t_k) + g_p(x_i, t_{k+1})\}, \quad i = 1, 2, \ldots, m \end{aligned} \tag{4.4}$$

where

$$g_p(x_i, t_{k+1}) = g(\tilde{V}_p(x_i, t_{k+1})) \quad \text{and} \quad \tilde{V}_p(x_i, t_{k+1}) = \sum_{i=1}^{m} \phi_{ij} C_j^p(t_{k+1}), \quad i = 1, 2, \ldots, n$$

Note that the problem of nonlinearity can also be avoided using extrapolated techniques.

In all calculations unless stated otherwise, we choose the initial condition

$$V(x, 0) = 4Hx(1 - x) \quad H \text{ is non zero and } a = \tfrac{1}{4}. \tag{4.5}$$

Before carrying out the numerical calculation of (1.4) using the predictor-corrector pair, let us determine the lengths $L_{\min}$ corresponding to $\alpha_{\min}$ (cf. (3.13)) for given values of a. Table 1 gives these $L_{\min}$ values which are all in the vicinity of the critical estimate π obtained in [6].

All the numerical results reported here are based on equally spaced collocation points $x_i = \frac{1}{4}i$ ($i = 1, 2, 3$) and it is seen in our numerical study that the significant eigenfunctions are ψ_1 and ψ_3 (Figs. 1–3).

(i) $L < L_{\min}$. In this case whatever the value of H, (i.e., however large the initial disturbance is) the solution $V(x, t)$ tends to zero as t increases (see Fig. 1).

(ii) $L_{\min} < L < L_B \equiv$ the length corresponding to $\alpha_B = 3.62759$. Let us take $L = L_1 (= 3.25661)$, the length corresponding to the solution of the form (1.7). The solutions corresponding to (3.11) are obtained for this case and both are solutions with a maximum. The respective maximum values are

$$H_1 = 0.43872 \quad \text{and} \quad H_2 = 0.14219.$$

Table 1

a	0.10	0.25	0.40	0.45	0.499
$L_{\min}$	3.15646	3.15339	3.14397	3.14220	3.14159

In fact, the solution corresponding to (1.7) is none other than the solution corresponding to the maximum H_2. However, in numerical calculations depending on the value of H in (4.5) we either get a solution corresponding to the solution with the maximum H_1 or the zero solution. For example, at $x = \frac{1}{2}$ for large t, $H = 0.15$ leads to $H_\infty = 0.42788$ (cf. H_1) and $H = 0.13$ leads to $H_\infty \approx 0$ (see Fig. 2). This means that the solution corresponding to H_1 is asymptotically stable while the other solution (corresponding to H_2) is not. Further, extensive numerical studies show that the solution given by (1.7) ($\bar{V}(x)$, say) behaves as a threshold curve in deciding which initial disturbance develops into a nonconstant steady state solution, i.e., if the initial data is such that

$$V(x, 0) \geqslant \bar{V}(x) \quad \text{for all } x,$$

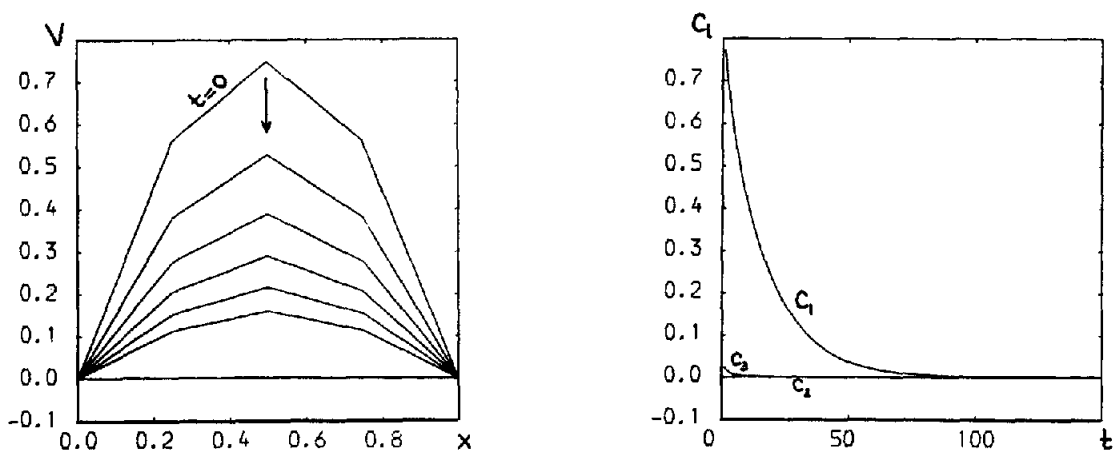

Fig. 1. $L < L_{\min}$; $L = 2.0$, $H = 0.75$, $V(x, t) \to 0$, for large t.

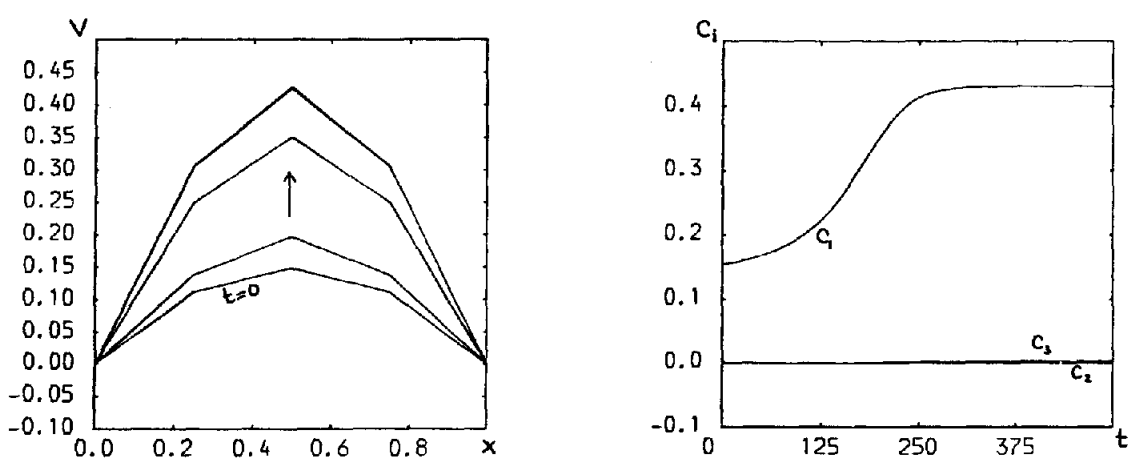

Fig. 2. $L_{\min} \leqslant L \leqslant L_B$; $L = 3.25661$, $H = 0.15$ leads to $H_\infty = 0.42788$. L_2-error $= 1.60469 \times 10^{-2}$.

the nonconstant solution corresponding to H_1 is obtained, while if

$$V(x,0) < \overline{V}(x) \quad \text{for all } x,$$

we get the zero solution.

(iii) $L > L_B$. We choose $L = 4.5$ and the corresponding solutions from (3.11) have a maximum $H_3 = 0.74603$ and a minimum $H_4 = -0.12547$. The numerical solutions tend to one or the other of these solutions depending on whether the initial disturbance is positive or negative (i.e., H is positive or negative). For example, $H = -0.2$ tends to $H_\infty = -0.12496$ (cf. H_4) (see Fig. 3).

These numerical results suggest that the bifurcation diagram of the problem will be as in Fig. 4.

When $\alpha < \alpha_{\min}$, the only solution is the zero solution. For α between $\alpha_{\min}$ and α_B depending on whether the initial disturbance is below or above the solution given by the branch PR, we get

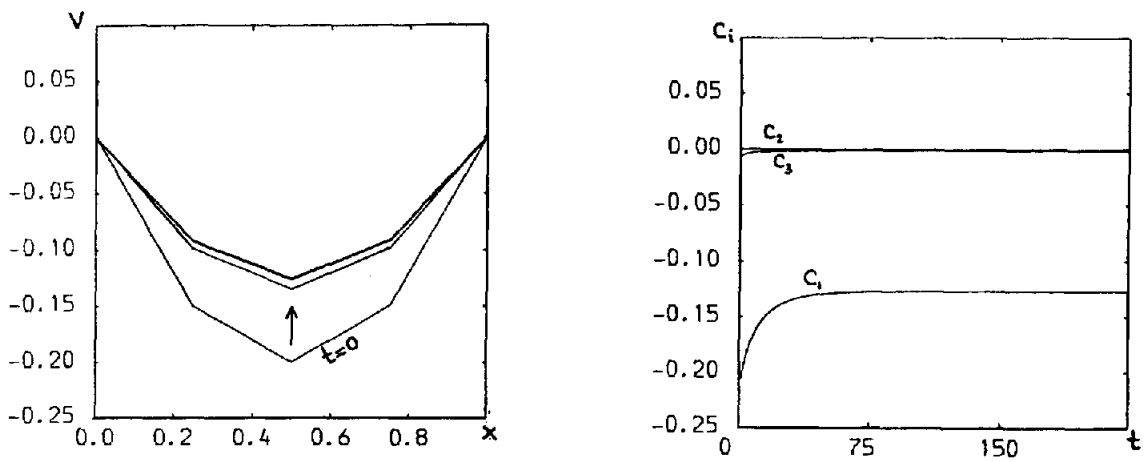

Fig. 3. $L > L_B$; $L = 4.5$, $H = -0.2$ leads to $H_\infty = -0.12496$. L_2-error $= 5.10660 \times 10^{-4}$.

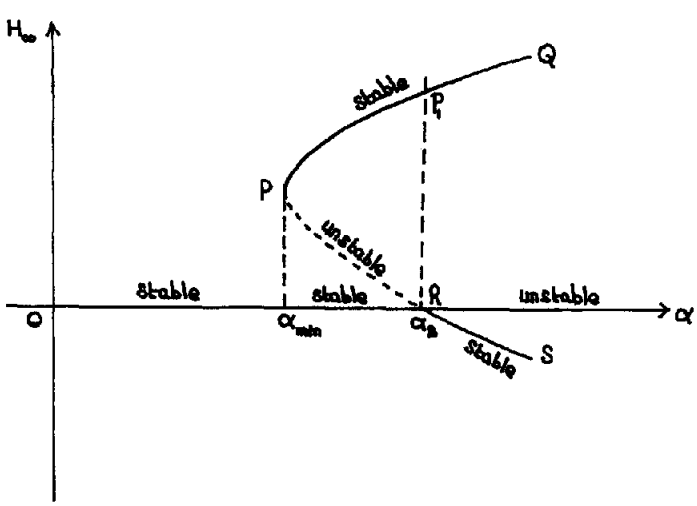

Fig. 4.

the zero solution or the solution corresponding to the branch PP_1. However, for $\alpha > \alpha_B$, the zero solution becomes unstable and we get the solutions corresponding to the branches P_1Q and RS depending on whether the initial disturbance is positive or negative.

The numerical computations were done on a DEC-10 computer by the FORTRAN IV programming language. When using the predictor–corrector formulation (4.3)–(4.4), at every time step the corrector was used only once in order to cut down the computational cost; because some results obtained by correcting upto convergence did not vary significantly with those obtained with one correction.

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